

Day 3: Data Visualization

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Date: 17-06-2026

Assignment:

Question 1: If you had to visualize the Titanic dataset, which chart would best show survival by gender and class — and why? 🧩 (Encourages you all to connect chart choice with storytelling.)

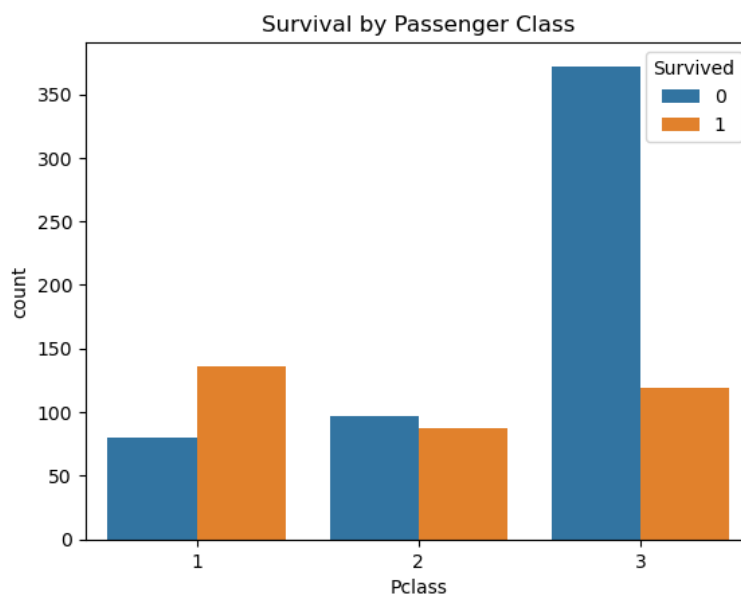
Answer:

A grouped bar chart would be the best choice because it allows us to compare survival counts across both gender and passenger class (Pclass) at the same time.

Example:

X-axis → Passenger Class (1st, 2nd, 3rd)

Bars → Male and Female survival counts in each class



0 -> Did not Survived, and 1-> Survived

This makes it easy to see that female passengers and first-class passengers had higher survival rates, helping tell the story clearly.

Question 2: How can poor design choices — like misleading scales or excessive color — distort the message of a visualization? 🧠 (Promotes critical thinking about ethics and clarity in data communication.)

Answer:

Poor design can make viewers draw incorrect conclusions.

Example 1: Misleading Scale

If a bar chart's Y-axis starts at 50 instead of 0, a small difference in survival rates may appear much larger than it actually is.

Example 2: Excessive Colors

Using too many bright colors in a survival chart can distract viewers and make important patterns harder to identify.

Example 3: Same Colors

If we use same colors on different categories we don't even understand which one belong from which class.

Conclusion:

Good visualizations should be clear, accurate, and easy to interpret, so the data tells the truth without misleading the audience.